

## NATCHEZ-ADAMS COUNTY AP, MS, USA

WMO: 722357

Lat: 31.617N

Lon: 91.300W

Elev: 83

StdP: 100.33

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03961

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -3.9       | -2.4 | -12.0                         | 1.3 | -0.8 | -8.8 | 1.8 | 2.1  | 9.6                   | 13.5 | 8.8 | 13.2 | 2.0                   | 10   | 0.382 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.8                    | 34.1            | 25.1 | 33.1 | 25.0 | 32.6 | 24.9 | 26.6                | 32.1 | 26.1 | 31.5 | 25.6 | 31.0 | 2.7                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.1                            | 20.4 | 29.2 | 24.3 | 19.4 | 28.8 | 24.0 | 19.0 | 28.6 | 83.3          | 31.8 | 80.9 | 31.4 | 79.0 | 31.5 | 28.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 8.3               | 7.3  | 6.4 | -7.4                       | 35.6 | 2.4                | 1.6 | -9.1                                               | 36.7 | -10.6      | 37.7 | -11.9      | 38.6 | -13.7      | 39.8 |
|                   |      |     | -8.2                       | 27.3 | 2.2                | 0.7 | -9.9                                               | 27.8 | -11.2      | 28.1 | -12.4      | 28.5 | -14.0      | 29.0 |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.7   | 8.9   | 11.3  | 14.6  | 18.4  | 22.8  | 26.0  | 27.2  | 27.1  | 24.4  | 19.1  | 13.6  | 10.3 |
|                                                                                  | DBStd       | 7.83   | 6.07  | 5.83  | 5.14  | 4.12  | 3.20  | 1.95  | 1.70  | 2.37  | 2.98  | 4.45  | 5.09  | 5.78 |
|                                                                                  | HDD10.0     | 261    | 97    | 50    | 18    | 1     | 0     | 0     | 0     | 1     | 0     | 1     | 23    | 70   |
|                                                                                  | HDD18.3     | 1159   | 298   | 206   | 134   | 49    | 7     | 0     | 0     | 2     | 2     | 47    | 157   | 257  |
|                                                                                  | CDD10.0     | 3429   | 62    | 87    | 162   | 254   | 397   | 481   | 533   | 529   | 433   | 282   | 131   | 78   |
|                                                                                  | CDD18.3     | 1286   | 4     | 9     | 20    | 52    | 146   | 231   | 275   | 272   | 186   | 70    | 14    | 7    |
|                                                                                  | CDH23.3     | 11581  | 7     | 36    | 111   | 389   | 1223  | 2169  | 2725  | 2682  | 1582  | 569   | 69    | 19   |
|                                                                                  | CDH26.7     | 4636   | 0     | 2     | 7     | 63    | 404   | 911   | 1228  | 1236  | 628   | 152   | 5     | 0    |
| Wind                                                                             | WSAvg       | 2.7    | 3.4   | 3.4   | 3.4   | 3.2   | 2.7   | 2.2   | 2.0   | 1.8   | 2.0   | 2.3   | 2.7   | 3.1  |
| Precipitation                                                                    | PrecAvg     | 1425   | 145   | 134   | 139   | 115   | 113   | 110   | 109   | 117   | 99    | 92    | 124   | 125  |
|                                                                                  | PrecMax     | 1890   | 321   | 291   | 336   | 299   | 259   | 288   | 184   | 301   | 350   | 243   | 285   | 217  |
|                                                                                  | PrecMin     | 1012   | 34    | 20    | 18    | 10    | 6     | 21    | 26    | 37    | 16    | 0     | 10    | 50   |
|                                                                                  | PrecStd     | 258    | 77    | 75    | 65    | 72    | 60    | 69    | 46    | 73    | 75    | 67    | 78    | 44   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 24.0  | 26.9  | 27.6  | 29.8  | 32.4  | 34.1  | 36.0  | 36.3  | 34.4  | 32.1  | 27.5  | 26.0 |
|                                                                                  |             | MCWB   | 19.0  | 19.7  | 19.7  | 22.3  | 24.0  | 24.7  | 25.6  | 25.1  | 24.2  | 22.9  | 21.0  | 20.1 |
|                                                                                  | 2%          | DB     | 22.4  | 24.1  | 26.2  | 28.0  | 31.3  | 33.0  | 34.0  | 34.2  | 32.8  | 29.9  | 25.9  | 23.0 |
|                                                                                  |             | MCWB   | 17.7  | 18.1  | 18.8  | 20.9  | 23.4  | 24.7  | 25.4  | 25.3  | 24.1  | 22.6  | 19.8  | 19.2 |
|                                                                                  | 5%          | DB     | 20.9  | 22.5  | 24.1  | 27.3  | 30.4  | 32.4  | 32.9  | 33.0  | 32.1  | 28.0  | 23.6  | 22.0 |
|                                                                                  |             | MCWB   | 16.8  | 17.5  | 17.8  | 20.2  | 23.0  | 24.7  | 25.3  | 25.3  | 24.0  | 21.1  | 18.6  | 18.4 |
|                                                                                  | 10%         | DB     | 18.6  | 21.0  | 22.7  | 26.1  | 28.9  | 31.3  | 32.4  | 32.4  | 30.8  | 27.1  | 22.2  | 19.9 |
|                                                                                  |             | MCWB   | 15.9  | 16.8  | 17.1  | 19.2  | 22.3  | 24.4  | 25.2  | 25.2  | 23.5  | 20.3  | 17.9  | 16.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 20.4  | 20.9  | 21.4  | 23.6  | 25.2  | 26.4  | 27.3  | 27.4  | 25.9  | 25.1  | 22.6  | 21.5 |
|                                                                                  |             | MCDB   | 23.2  | 24.6  | 25.5  | 28.1  | 30.8  | 32.1  | 32.5  | 32.7  | 31.1  | 29.5  | 25.5  | 24.1 |
|                                                                                  | 2%          | WB     | 18.7  | 19.7  | 20.2  | 22.3  | 24.5  | 25.8  | 26.5  | 26.7  | 25.4  | 23.9  | 21.2  | 20.0 |
|                                                                                  |             | MCDB   | 21.6  | 23.0  | 24.2  | 26.7  | 29.6  | 31.2  | 32.1  | 32.2  | 30.6  | 27.9  | 24.0  | 22.9 |
|                                                                                  | 5%          | WB     | 17.4  | 18.6  | 19.3  | 21.5  | 23.8  | 25.4  | 26.0  | 26.1  | 24.8  | 23.0  | 19.8  | 18.8 |
|                                                                                  |             | MCDB   | 19.2  | 21.5  | 23.2  | 25.6  | 28.6  | 30.7  | 31.6  | 31.6  | 29.8  | 26.4  | 23.1  | 21.4 |
|                                                                                  | 10%         | WB     | 15.8  | 17.1  | 18.4  | 20.6  | 23.1  | 24.9  | 25.6  | 25.6  | 24.1  | 21.8  | 18.4  | 17.2 |
|                                                                                  |             | MCDB   | 18.5  | 20.1  | 22.1  | 24.5  | 27.6  | 30.0  | 31.0  | 30.9  | 28.9  | 25.1  | 21.0  | 19.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 10.8  | 11.0  | 11.4  | 11.8  | 11.0  | 10.0  | 9.8   | 10.1  | 10.9  | 12.2  | 12.0  | 10.7 |
|                                                                                  |             | MCDBR  | 10.7  | 11.4  | 11.9  | 11.7  | 11.5  | 10.8  | 10.6  | 11.0  | 11.9  | 12.4  | 12.4  | 11.0 |
|                                                                                  |             | MCWBR  | 7.5   | 7.1   | 6.8   | 6.3   | 5.2   | 4.7   | 4.3   | 4.4   | 4.8   | 6.3   | 7.7   | 7.6  |
|                                                                                  | 5% WB       | MCDBR  | 9.6   | 10.3  | 10.4  | 9.9   | 10.0  | 9.7   | 9.7   | 9.9   | 9.9   | 9.7   | 10.4  | 10.0 |
| MCWBR                                                                            |             | 8.2    | 7.4   | 6.9   | 6.0   | 5.0   | 4.6   | 4.3   | 4.4   | 4.5   | 5.4   | 7.5   | 7.6   |      |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.324  | 0.342 | 0.373 | 0.408 | 0.440 | 0.474 | 0.500 | 0.496 | 0.441 | 0.370 | 0.347 | 0.330 |      |
|                                                                                  | taud        | 2.536  | 2.483 | 2.392 | 2.296 | 2.252 | 2.199 | 2.152 | 2.161 | 2.294 | 2.457 | 2.477 | 2.525 |      |
|                                                                                  | Ebn at Noon | 904    | 918   | 908   | 885   | 854   | 821   | 798   | 796   | 830   | 874   | 866   | 876   |      |
|                                                                                  | Edh at Noon | 85     | 98    | 115   | 132   | 139   | 146   | 152   | 149   | 126   | 100   | 89    | 82    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.76   | 3.25  | 4.43  | 5.54  | 6.11  | 6.25  | 6.10  | 5.73  | 5.07  | 4.29  | 3.19  | 2.45  |      |
|                                                                                  | RadStd      | 0.23   | 0.52  | 0.36  | 0.41  | 0.37  | 0.50  | 0.37  | 0.46  | 0.42  | 0.56  | 0.33  | 0.27  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (24 neighbors) | +0.33 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +123    | +80     |

Nomenclature: See separate page